

(1) Data layer — vintage-aware collectors

exchange

macro (available_at)

alternative

*eval. archive:
3 bands*

(2) Store layer — embedded analytical stores

raw multi-band history
(timestamps / vintages)

merged market panels

analytics: readiness,
WMI / ACWMI, snapshots

(3) Logic layer — compilation $\Pi_t = B_t \circ M_t \circ A_t$

BandPIT clock
decision_asof = (t-1) 23:59
(no same-day look-ahead)

honesty / missingness gates
 $B, U, H \rightarrow \text{WMI} = BUH$
ACWMI (regime-cond.)

availability shocks O_t
(quasi-exogenous
thickness lever)

(4) Service layer — world bundle contract (read API)

completeness · honesty B, U, H · WMI / ACWMI · should_ai_abstain · thin_world
content: macro_tilt, alt_tilt, regime · audit: evidence_ids

(no feed)

(5) Consumer layer — frozen prompts, temp. 0 (GPT / DeepSeek / GLM / Gemini)

COMPILED
full bundle +
hard should_ai_abstain

UNGATED
same content;
no hard flag

RAW
mom5 only;
no world model

BLIND
direct ask;
no data feed

Live abstention outcomes: Sec.~Experiments (Table / four-arm figure).